

Utilization of mango peels as a source of pectin and polyphenolics

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Dedicated to Professor Ronald E. Wrolstad, Corvallis/OR, USA

Abstract

Two different options for the combined recovery of pectin and phenolic compounds from mango peels, a byproduct of industrial mango processing, were developed. After extraction of dried mango peels with diluted sulfuric acid, the phenolic compounds were adsorbed using a styrene–divinylbenzene copolymerisate resin, and pectin was obtained from the effluent by precipitation with ethanol. Phenolic compounds were recovered from the resin with methanol and the eluate was lyophilized (Process I). Alternatively, the pectin was precipitated by adding the crude extract to ethanol. After removal of the organic solvent, the phenolic compounds were obtained from the aqueous phase of the precipitation bath using the adsorbent resin as described before (Process II). While in total, 129.4 mg/g polyphenols were detected in the lyophilizate obtained from Process I, only 71.0 mg/g dm could be recovered from Process II. The profiles of the polyphenols were almost identical, revealing that during pectin precipitation preferential adsorption of polyphenolic compounds to the pectin may be excluded. Besides the characterization of the pectins and the phenolic compounds, investigations into the influence of the drying temperature on the polyphenolic content of the peels were carried out, indicating a significant loss of flavonol glycosides depending on heat exposure. On the other hand, some xanthone glycosides were formed during the drying process. Furthermore, antioxidative capacities of the lyophilized eluates were investigated using the DPPH, TEAC and FRAP assays. The antioxidative capacity of the extracts exceeded that of mangiferin and quercetin 3-*O*-glucoside, respectively, thus demonstrating mango peels to be a suitable source of health-beneficial compounds. The lyophilizates obtained from Process I showed higher antioxidative capacities in all three assays. These findings indicate a correlation between the amount of phenolic compounds and the antioxidative capacity.

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Keywords: *Mangifera indica* (L.); Peels; Flavonols; Xanthenes; Pectin; Adsorption; Antioxidant capacity

Industrial relevance: Byproducts of mango processing amount to 35–60% of the total fruit weight. Their complete exploitation for further product recovery is a promising measure from both an environmental and economic point of view. In our previous study mango peels were found to be a rich source of pectin, with a high degree of esterification and phenolic compounds, like flavonol *O*- and xanthone *C*-glycosides. Therefore, two alternative processes for the combined recovery of pectin and polyphenols, which can easily be integrated in an existing pectin production process, were developed in the present study.

1. Introduction

With a global production exceeding 26 million tons in 2004 (FAOSTAT, 2005), mangoes (*Mangifera indica* L., Anacardiaceae) are one of the most important tropical fruits. Besides the fresh fruit, processed mango products such as juices, nectars, concentrates, jams, jelly powders, fruit bars,

flakes and dried fruits have become increasingly popular in Europe (Loeillet, 1994; Mahayothee, 2005; Pott, Neidhart, Mühlbauer, & Carle, 2005; Singh, Dhuique-Mayer, & Lozano, 2000; Vásquez-Cañedo, Neidhart, & Carle, 2004). The export of mango pulp as a raw material for these products was estimated to be about 62% of total exports of fruits and vegetables in India, the world's largest mango producer (Sreenath, Krishna, & Santhanam, 1995). The edible pulp makes up 33–85% of the fresh fruit, while the peel and the kernel amount to 7–24% and 9–40%, respectively (Wu, Chen, & Fang, 1993). According to Larrauri, Rupérez,

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Borroto, and Saura-Calixto (1996), byproducts of industrial mango processing may amount to 35–60% of the total fruit weight.

Because these byproducts represent a serious disposal problem, ways for a sustainable agricultural production have been searched. While the utilization of the mango kernels as a source of fat, natural antioxidants, starch, flour and feed, respectively, has extensively been investigated (Arogba, 2002; Kaur, Singh, Sandhu, & Guraya, 2004; Moharram & Moustafa, 1982; Puravankara, Boghra, & Sharma, 2000; Ravindran & Sivakanesan, 1996), studies on peels are scarce. Their use for the production of biogas (Madhukara, Nand, Raju, & Srilatha, 1993; Mahadevaswamy & Venkataraman, 1990) or dietary fiber with a high antioxidant activity (Larrauri, Goñi, Martín-Carrón, Rupérez, & Saura-Calixto, 1996; Larrauri, Rupérez et al., 1996; Larrauri, Rupérez, & Saura-Calixto, 1997) has been described in the past.

Most studies on the exploitation of mango peels have been dealing with their use as a source of pectin, which is considered a high quality dietary fiber (Beerh, Raghuramaiah, & Krishnamurthy, 1976; Pedroza-Islas & Aguilar-Esperanza, 1994; Srirangarajan & Shrikhande, 1976; Tandon & Garg, 1999; Tandon, Kalrat, Singh, & Garg, 1991). A screening of 14 mango cultivars recently carried out in our laboratory has demonstrated the content and the degree of esterification of mango peel pectins to range from 12% to 21% and from 56% to 66%, respectively (Berardini et al., 2005). Furthermore, mango peels have been shown to be a rich source of flavonol *O*- and xanthone *C*-glycosides (Berardini et al., 2005; Schieber, Berardini, & Carle, 2003), gallotannins and benzophenone derivatives (Berardini, Carle, & Schieber, 2004).

In continuation of our studies on the recovery of valuable compounds from byproducts of fruit and vegetable processing (Kammerer, Claus, Schieber, & Carle, 2005; Kammerer, Gajdos Kljusuric, Carle, & Schieber, 2005; Schieber, Hilt et al., 2003; Stoll, Schweiggert, Schieber, & Carle, 2003), two processes for the combined recovery of pectin and polyphenolics from mango peels using adsorption technology were developed in the present study. The processes were evaluated by characterization and quantification of the pectins and phenolic compounds in the peels and in the polyphenol-enriched eluates. Furthermore, the eluates were screened for their antioxidant activity using the DPPH, TEAC and FRAP assays, since various health-promoting effects of phenolic compounds have mainly been ascribed to their antioxidant activity. The chemistry behind the most common antioxidant activity assays has recently been reviewed in detail (Huang, Ou, & Prior, 2005). After mango processing the peels have to be dried immediately because they are prone to rapid enzymatic pectin degradation and microbial spoilage. Therefore, the effects of different time–temperature regimes applied during drying on the stability of phenolic compounds of the peels were also investigated.

2. Materials and methods

2.1. Chemicals and solvents

Reference compounds used for identification and quantification purposes with HPLC and HPLC-MS were as follows: ellagic acid (Serva, Heidelberg, Germany); quercetin 3-*O*-xyloside, quercetin 3-*O*-arabinofuranoside (Plantech, Reading, UK); quercetin, quercetin 3-*O*-arabinopyranoside, quercetin 3-*O*-arabinoglucoside (Roth, Karlsruhe, Germany); mangiferin [2-*C*- β -D-glucopyranosyl-1,3,6,7-tetrahydroxy-xanthone], quercetin 3-*O*-galactoside, quercetin 3-*O*-glucoside, quercetin 3-*O*-rhamnoside, kaempferol 3-*O*-glucoside, isorhamnetin 3-*O*-glucoside (Extrasynthèse, Lyon, France). Due to the limited amounts available, quercetin 3-*O*-arabinofuranoside and -pyranoside could only be used for identification purposes. Gallic acid, DPPH [2,2-diphenyl-1-picrylhydrazyl], ABTS [2,2'-azino-bis(3-ethylbenzthiazoline-6-sulfonic acid)], ABAP [2,2'-azobis(2-amidinopropane) dihydrochloride], TPTZ [2,4,6-tripyridyl-*s*-triazine], and Trolox [(+/-)-6-hydroxy-2,5,7,8-tetramethyl-chroman-2-carboxylic acid] were obtained from Sigma (St. Louis, USA). All other chemicals and solvents were of analytical or gradient grade and supplied by VWR (Darmstadt, Germany).

2.2. Plant material

Mango peels were obtained from mango purée production on pilot plant-scale at Hohenheim University. Mature Brazilian mango fruits of the cultivar Tommy Atkins were purchased from the local market. After blanching of the fruits for 7 min using hot steam, the peels were removed from the flesh with a stainless steel knife and stored in vacuum-sealed plastic bags at -20°C until use.

2.3. Drying of the peels

Mango peels were dried in a circulating air drying oven at 50°C for 270 min, at 70°C for 180 min, and at 90°C for 90 min, respectively, until a drying loss of 82%. For comparison, an aliquot of the peels was lyophilized. All samples were analyzed for their contents of xanthone and flavonol glycosides as described below.

2.4. Extraction of pectin from mango peels

Mango peels dried at 90°C were manually ground. Batches of 666 g of the peels were mixed with 20 L of deionized water in a 30 L pilot plant-scale reaction vessel (Type EL 10, Schweizer, Riehen, Switzerland). The mixture was heated at 100°C under stirring and subsequently cooled. After adjusting the pH to 1.5 using sulfuric acid (98%), the solution was heated at 90°C for 2.5 h. During the last 60 min of heating, vacuum was used to reduce the volume of the solution by 50%. The solution was cooled and filtered through a perlon filter cloth using a small-scale press (Fischer

and Co. KG, Neuss, Germany). After clarifying with a settling filter including a deep bed filter (SeitzSchenk Filtersystems, Waldstetten, Germany), the solution was kept at $-20\text{ }^{\circ}\text{C}$ until use.

2.5. Recovery of pectin and phenolic compounds

Amberlite XAD 16 HP (Rohm and Haas, Darmstadt, Germany) was slurried in water for 2 h before use. Five hundred milliliters of the resin were filled into a XK 50/100 glass column ($90 \times 5\text{ cm i.d.}$, Amersham Pharmacia Biotech, Uppsala, Sweden) and washed with five bed volumes of distilled water to remove impurities using a Mohno pump (Netzsch, Waldkraiburg, Germany). The temperature of the column was kept at $60\text{ }^{\circ}\text{C}$ during the whole process. Fig. 1 shows the two different processes for the combined recovery of pectin and phenolic compounds which are characterized as follows:

2.5.1. Process I

The mango peel extract was applied to the column at a flow rate of 10 bed volumes per hour. Since preliminary laboratory-scale experiments had shown optimum adsorption behavior of the resin when up to 40 bed volumes were used, 20 L of the mango peel extract were used for each pilot plant experiment. The effluent was collected for pectin recovery as described below, and residues of pectin were removed by washing the column with 2 bed volumes of water. Phenolic

compounds were desorbed using 4 bed volumes of methanol. The alcohol was evaporated in vacuo. Subsequently, the remaining aqueous solution was lyophilized and the residue stored in amber glass bottles in a desiccator until analyzed.

2.5.2. Process II

In a second experiment, 6 L aliquots of the mango peel extract were added to 12 L of ethanol (96%). After 30 min, the precipitated pectin was recovered by filtration using a perlon filter cloth, washed with ethanol, pressed and finally lyophilized. Twelve liters of the alcoholic filtrate were evaporated in vacuo at $30\text{ }^{\circ}\text{C}$ to yield 1 L of an aqueous solution, which was applied to the column at a flow rate of 10 bed volumes per hour. Subsequently, the column was washed with two bed volumes of water. The effluent during the adsorption and washing process was discarded. Phenolic compounds were desorbed using four bed volumes of methanol. The alcoholic eluate was treated as described under 'Process I'.

2.6. Characterization of pectin and phenolic compounds

2.6.1. Color of the pectin extracts

Color parameters (L^* , a^* , b^*) of the mango peel extract before and after adsorptive removal of phenolic compounds were recorded with a Lambda 20 photometer equipped with an RSA-PE-20 integration sphere and using WinCol software (Version 2.05) (Perkin Elmer, Dreieich, Germany) using

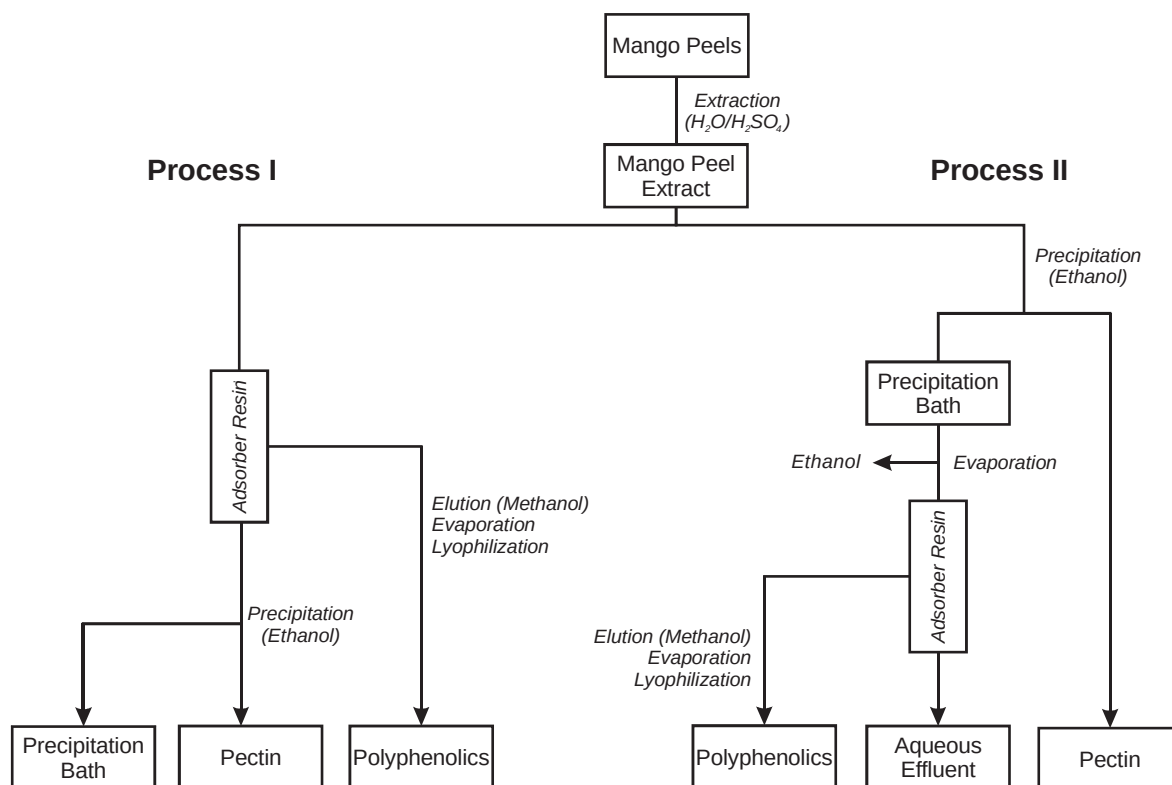


Fig. 1. Processes I and II for the combined recovery of pectin and phenolic compounds from mango peels.

illuminant D65 and 10° observer angle. Chroma (C^*) and hue angle (h°) were calculated as follows:

$$C^* = \sqrt{a^{*2} + b^{*2}}; h^\circ = \arctan\left(\frac{b^*}{a^*}\right).$$

2.6.2. Yield and color characteristics of pectin

Two hundred fifty milliliters of the mango peel extract and the same volume of pectin containing effluent obtained from Process I were added to 500 mL of ethanol. After 30 min, the precipitate was passed through a perlon filter cloth, washed with 1 L of ethanol, manually pressed and lyophilized. Pectin yields were determined gravimetrically. After grinding the samples with a water-cooled YellowLine A10 laboratory mill (Ika-Werke, Staufen, Germany), color parameters (L^* , a^* , b^* , C^* and h°) were determined using an A5 Chroma-Meter CR 300 (Minolta, Langenhagen, Germany). The samples were stored in a desiccator until further analyses.

2.6.3. Extraction and determination of xanthone C- and flavonol 3-O-glycosides

For the determination of the phenolic contents, the dried or lyophilized mango peels were powdered using liquid nitrogen and a stainless steel Waring Blendor as described by Rodriguez-Saona and Wrolstad (2001). Aliquots of 2.5 g were weighed into amber glass round-bottomed flasks. After addition of 0.5 g of ascorbic acid, the flasks were flushed with nitrogen and the mixture was extracted with 50 mL of aqueous acetone (80%, v/v) for 3 h under stirring. The extract was centrifuged (10 min, 3480 $\times g$), and the residue was re-extracted with 50 mL of the same solvent for 10 min. The organic solvent was removed from the combined supernatants by evaporation in vacuo at 30 °C. The aqueous solution was transferred into a graduated flask and made up to 50 mL with deionized water. After filtration, aliquots of 20 mL were used for further purification. For the quantification of the phenolic compounds in the mango peel extract used in Processes I and II, aliquots of 10 mL of the solution were used. The polyphenolics in the alcoholic supernatant obtained from Process II were determined by evaporating 200 mL of the solution in vacuo until the organic solvent was removed. The aqueous solution was transferred into a graduated flask and made up to 50 mL with deionized water. After filtration, aliquots of 20 mL were used for further sample clean-up. Phenolic compounds in the precipitated pectin were quantified by weighing 1 g of the lyophilized sample into an amber glass Erlenmeyer flask and adding 100 mL of acidified water (sulfuric acid, pH 1.5). After flushing with nitrogen, the flask was sealed with parafilm and the solution was stirred for approximately 2 h until the pectin was completely dissolved. This solution was directly used for polyamide clean-up.

Polyamide CC6 (2 g, particle size 0.05–0.16 mm; Macherey-Nagel, Düren, Germany) was filled into an Econo-Pac column (BioRad, Munich, Germany) and successively conditioned with 25 mL of methanol and 50 mL of deionized

water before the samples were applied to the column. After washing with water (50 mL), the polyphenolic fraction was recovered by elution with methanol (100 mL). The eluate was evaporated to dryness, and the residue was dissolved in 0.5 mL of methanol. The solution was membrane-filtered (0.45 μ m, Whatman Inc., Clifton, USA) and used for HPLC analyses. For the quantification of the phenolic compounds in the eluates obtained in Processes I and II, the lyophilizates were ground in a mortar. 10 mg of the samples were dissolved in 1 mL of methanol, membrane-filtered (0.45 μ m, Whatman Inc., Clifton, USA) and directly used for HPLC analyses.

The separation of phenolic compounds was performed using an Agilent HPLC series 1100 system (Agilent, Waldbronn, Germany) equipped with ChemStation software, a model G1322A degasser, a model G1312A binary gradient pump, a model G1329/1330A thermoautosampler, a model G1316A column oven, and a model G1315B diode array detector. The column used was a 4 μ m Synergi Hydro-RP (150 $\times$ 3.0 mm i.d.; Phenomenex, Torrance, USA), with a C18 ODS guard column (4.0 $\times$ 2.0 mm i.d.), which was operated at a temperature of 25 °C. The mobile phase consisted of 2% (v/v) acetic acid in water (eluent A) and of 0.5% acetic acid in water and acetonitrile (50:50, v/v; eluent B). The gradient program was as follows: 0% B to 25% B (15 min), 25% B to 30% B (35 min), 30% B to 80% B (10 min), 80% B to 0% B (0.5 min). The injection volume for all samples was 4 μ L. Simultaneous monitoring was performed at 320 nm (xanthenes), and 370 nm (flavonols) at a flow rate of 0.6 mL/min. UV/Vis-spectra were recorded from 200 to 600 nm (peak width 0.2 min).

LC-MS analyses were performed with the same HPLC system as described above connected in series with a Bruker (Bremen, Germany) model Esquire 3000+ ion trap mass spectrometer fitted with an ESI source. Negative ion mass spectra of the column eluate were recorded in the range m/z 50–1000. Nitrogen was used as the drying gas at a flow rate of 10.0 L/min and at a pressure of 4.14 bar. The nebulizer temperature was set at 365 °C.

Individual compounds were identified by their retention times, UV/Vis- and mass spectra (Schieber, Berardini et al., 2003) and quantified using a calibration curve of the corresponding standard compound. When reference compounds were not available, the calibration curve of structurally related substances was used including a molecular weight correction factor (Chandra, Rana, & Li, 2001). All determinations were performed in duplicate.

2.6.4. Quantification of hydrolyzable tannins

The quantification of hydrolyzable tannins was carried out spectrophotometrically according to the method described by Inoue and Hagerman (1988). Aliquots of 400 mg of the lyophilizates were sonicated in 1 mL of aqueous acetone (70%, v/v) for 30 min. The extracts were vacuum filtered through an Econo-Pac column (BioRad, Munich, Germany) filled with glass wool into a screw cap test tube. The column was washed with 5 mL of 2 N H₂SO₄. The test

tube was flushed with nitrogen and sealed tightly. After heating at 100 °C for 26 h, the samples were made up to 50 mL with water. An aliquot of 1 mL was pipetted into a 25 mL graduated flask, and 1.5 mL of a 0.667% methanolic rhodanine solution was added. After 5 min, 1 mL of 0.5 N aqueous potassium hydroxide solution was added. After 2.5 min, the graduated flask was made up with water, and after another 10 min the absorbance at 520 nm was recorded using a Lambda 20 spectrometer (Perkin Elmer, Überlingen, Germany). The amount of non-esterified gallic acid was determined in the same way omitting the hydrolysis step.

2.7. Determination of the antioxidant capacity

The antioxidant capacity of the lyophilized eluates obtained from Processes I and II and of the reference compounds gallic acid, mangiferin and quercetin 3-*O*-glucoside was determined by the DPPH, TEAC and FRAP assays. The samples were dissolved in methanol (1 g/L). For calibration, Trolox was used as a reference in all assays. The antioxidant capacity of the samples was calculated as μmol Trolox per 100 g of sample. All determinations were performed using a Cary 100 Cone spectrophotometer (Varian, Palo Alto, CA, USA).

2.7.1. DPPH assay

In a cuvette, 1.95 mL of a methanolic solution of DPPH (6×10^{-5} mol/L) was mixed with 50 μL of the sample solution and 50 μL of water. The absorbance was recorded continuously at 515 nm until the reaction reached a plateau ($\Delta E \leq 0.003 \text{ min}^{-1}$) (Bandonienė, Murkovic, Pfannhauser, Venskutonis, & Gruzdienė, 2002; Brand-Williams, Cuvelier, & Berset, 1995; Schlesier, Harwat, Böhm, & Bitsch, 2002; Von Gadow, Joubert, & Hansmann, 1997).

2.7.2. TEAC assay

A phosphate buffer (pH 7.4) was prepared by mixing 818 mL of a 66 mmol/L Na_2HPO_4 solution with 182 mL of a 66

mmol/L KH_2PO_4 solution and 150 mmol sodium chloride. For preparing the radical solution, 0.5 mL of a 20 mmol/L solution of ABTS in the phosphate buffer was mixed with 100 mL of a 2.5 mmol/L solution of ABAP in the phosphate buffer and heated at 60 °C for 15 min in a water bath. For the determination of the antioxidant capacity, 40 μL of the sample solution and 1.96 mL of the radical solution were mixed in a cuvette. After 6 min, the absorption was measured at 734 nm (Van den Berg, Haenen, Van den Berg, & Bast, 1999; Van den Berg, Haenen, Van den Berg, Van der Vijgh, & Bast, 2000).

2.7.3. FRAP assay

The FRAP reagent was prepared by mixing 2.5 mL of a 10 mmol/L TPTZ solution in 40 mmol/L hydrochloric acid with 2.5 mL of a 20 mmol/L FeCl_3 solution and 25 mL of a 0.3 mol/L acetate buffer (pH 3.6). For the determination of the antioxidant capacity, 50 μL of the sample solution, 150 μL of water and 1.5 mL of the FRAP reagent were mixed in a cuvette. After 4 min, the absorption was measured at 593 nm (Benzie & Strain, 1996; Luximon-Ramma, Bahorun, Soobratte, & Aruoma, 2002; Pulido, Bravo, & Saura-Calixto, 2000; Schlesier et al., 2002).

3. Results and discussion

A screening of 14 mango cultivars recently carried out had demonstrated the peels to be a rich source of flavonol *O*- and xanthone *C*-glycosides. In this context, the cultivar ‘Tommy Atkins’ was most promising because its peels showed the highest contents of polyphenolics and also contained high quality pectin (Berardini et al., 2005). Consequently, in the present study Tommy Atkins peels were used for the recovery of valuable compounds.

Contents of polyphenolic compounds in mango peels dried at 90 °C for 90 min are shown in Table 1. Total amounts of 4066 mg/kg dry matter (dm) were consistent with our previous findings (Berardini et al., 2005), however,

Table 1

Contents (calculated as mg/kg dm of peels) of phenolic compounds at different stages of the process ($n=2$)

	Dried mango peels (90 °C, 1 h 30 min)	Mango peel extract	Precipitation bath
Mangiferin	1690.4 $\pm$ 134.8	1190.9 $\pm$ 22.1	883.3 $\pm$ 49.2
Isomangiferin ¹	134.5 $\pm$ 1.9	51.9 $\pm$ 0.2	33.8 $\pm$ 4.6
Mangiferin gallate ²	321.9 $\pm$ 24.4	97.7 $\pm$ 0.8	74.3 $\pm$ 5.1
Isomangiferin gallate ²	82.0 $\pm$ 7.8	19.7 $\pm$ 0.2	19.4 $\pm$ 3.8
Quercetin 3- <i>O</i> -galactoside	651.2 $\pm$ 44.1	n.d.	n.d.
Quercetin 3- <i>O</i> -glucoside	557.7 $\pm$ 39.7	31.0 $\pm$ 0.7	22.7 $\pm$ 2.0
Quercetin 3- <i>O</i> -xyloside	207.3 $\pm$ 14.3	n.d.	n.d.
Quercetin 3- <i>O</i> -arabinopyranoside ³	101.5 $\pm$ 10.0	n.d.	n.d.
Quercetin 3- <i>O</i> -arabinofuranoside ³	103.6 $\pm$ 8.0	n.d.	n.d.
Quercetin 3- <i>O</i> -rhamnoside	20.1 $\pm$ 1.5	n.d.	n.d.
Kaempferol 3- <i>O</i> -glucoside	36.0 $\pm$ 2.2	n.d.	n.d.
Rhamnetin 3- <i>O</i> -galactoside/glucoside ⁴	94.4 $\pm$ 6.3	4.3 $\pm$ 0.0	4.3 $\pm$ 1.2
Quercetin	65.3 $\pm$ 7.1	23.4 $\pm$ 1.1	26.8 $\pm$ 7.3
Ellagic acid	n.d.	Traces	17.6 $\pm$ 0
Total	4066.0 $\pm$ 151.1	1418.8 $\pm$ 22.1	1012.5 $\pm$ 50.4

Quantified as ¹mangiferin ²mangiferin (including molecular weight correction factor) ³quercetin 3-*O*-xyloside ⁴isorhamnetin 3-*O*-glucoside; n.d.: not detected.

significant differences were observed as regards the contents of individual compounds. Whereas in the cultivar screening the two main compounds mangiferin and quercetin 3-*O*-galactoside were detected in almost the same amounts (1263.2 and 1217.3 mg/kg, respectively) and the xanthone glycosides represented about 31.6% of the total contents of flavonoid and xanthone glycosides, in the present study mangiferin contents (1690.4 mg/kg) were 2.6 times higher than those of quercetin 3-*O*-galactoside. The xanthone glycosides represented 54.8% of the total contents of flavonol and xanthone glycosides. This variability may be ascribed to factors such as soil conditions, exposition to UV radiation, ripening stage, and phytosanitary status. Furthermore, storage conditions may also affect polyphenol contents.

If mango peels are to be used as a raw material for the recovery of functional compounds, they have to be dried immediately to prevent the peels from enzymatic pectin degradation and microbial spoilage. Phenolic compounds are known to be heat-sensitive. Therefore, the impact of the time–temperature regime during drying on the contents of individual phenolics was investigated. The results were recalculated based on the dry matter content of the lyophilized control sample (13% dm).

The hot-air dried peels exhibited similar total amounts compared to the control sample (Fig. 2), with reductions ranging from 1% (50 °C) to 6% (70 °C). However, significant changes of individual compounds during the drying process were observed. While mangiferin contents slightly decreased at elevated temperatures, the contents of the other xanthone derivatives significantly increased, and this effect was even more pronounced with a rise in temperature (Fig. 3). Iso-mangiferin gallate contents increased by 94% when the peels

were dried at 90 °C, while isomangiferin and mangiferin gallate showed amounts which were 95 and 104% higher compared to the lyophilized control sample. The observed changes may be attributed to the formation of xanthenes from benzophenone derivatives, which were recently identified in mango peels (Berardini et al., 2004) and which are considered precursors of xanthone *C*-glycosides (Tanaka, Sueyasu, Nonaka, & Nishioka, 1984). For the formation of xanthone compounds oxidative conditions are required, which are given during the circulating air drying process.

In contrast, a different behavior was observed for the quercetin glycosides. The effect of the time–temperature regime on their stability is shown in Fig. 4. Quercetin 3-*O*-galactoside showed a distinct degradation of up to 35% of the value of the control sample with higher drying temperatures. With a maximum degradation of 10% at 90 °C, quercetin 3-*O*-glucoside was not as strongly affected as the galactoside. In contrast, the amount of the other quercetin glycosides was comparatively constant and did not show significant changes during any of the drying processes. The amount of the quercetin aglycone increased with higher temperatures, due to hydrolysis of individual quercetin glycosides. Kaempferol 3-*O*-glycoside decreased up to 10% with higher temperatures, while no distinct trend was observable for the rhamnetin hexosides (data not shown). Despite shorter drying times at higher temperatures the loss of flavonol glycosides was more pronounced.

Contents of phenolic compounds determined in the acidic mango peel extract are listed in Table 1. The results are calculated on a dry matter basis of the mango peels. Most strikingly, only 35% of the total contents found in the dried peels were detected in the crude extract. While the xanthone glycosides were still present in appreciable amounts, the

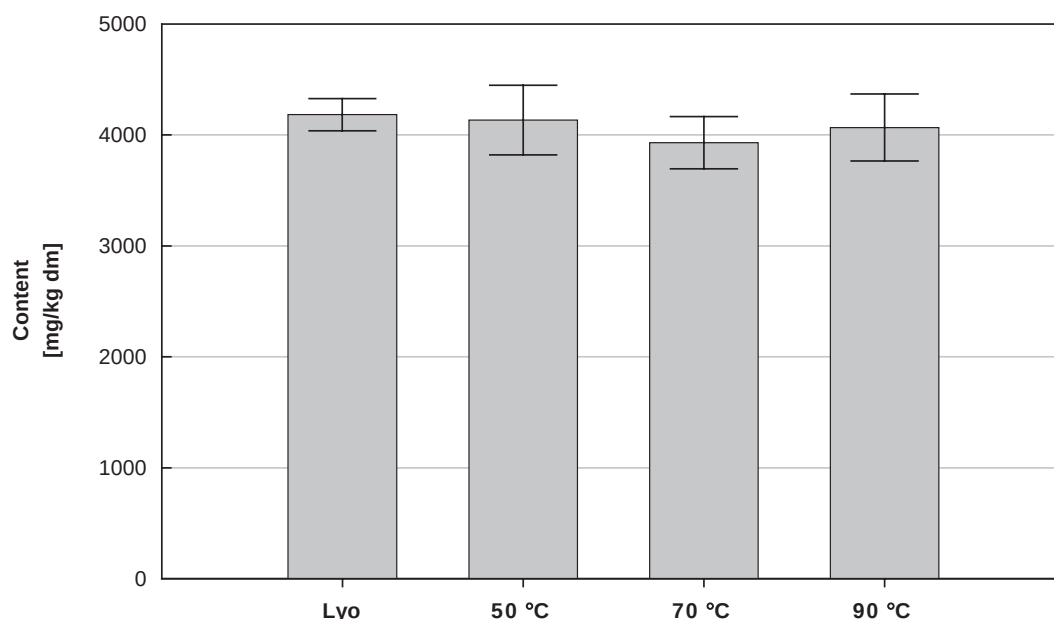


Fig. 2. Total contents of flavonol 3-*O*- and xanthone *C*-glycosides in mango peels dried at 50, 70 and 90 °C, respectively. Lyo: lyophilized sample.

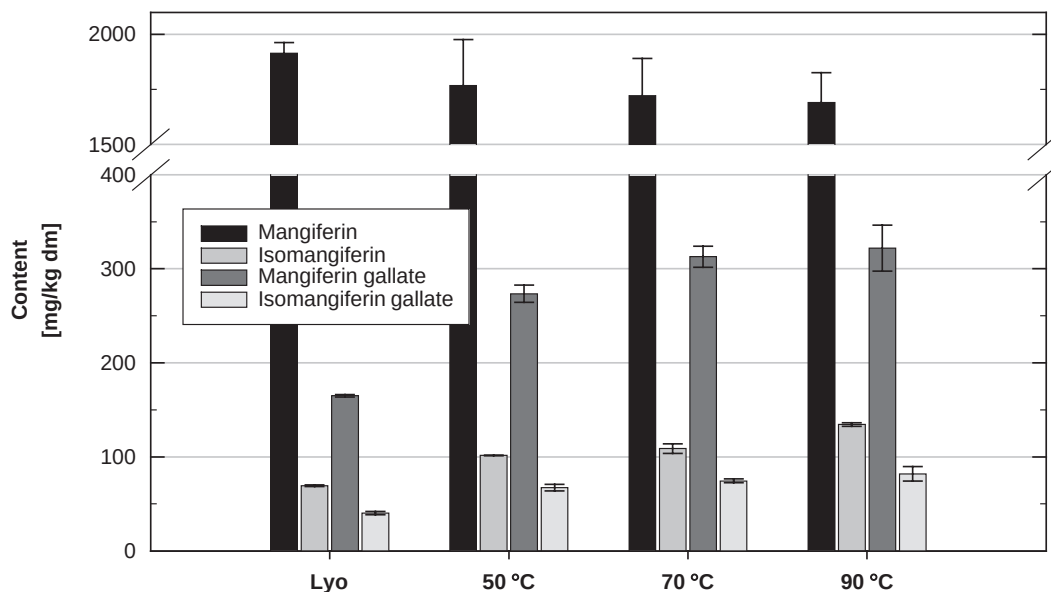


Fig. 3. Contents of individual xanthone C-glycosides in mango peels dried at 50, 70 and 90 °C, respectively. Lyo: lyophilized sample.

flavonol glycosides were almost completely degraded, with only quercetin 3-*O*-glucoside and rhamnetin glycosides being detectable. The partial hydrolysis of flavonol *O*-glycosides may be attributed to the acidic conditions applied during extraction of pectin at an elevated temperature. This assumption is corroborated by the fact that the quercetin aglycone was still detectable and xanthone C-glycosides were affected to a much lesser extent. Interestingly, traces of ellagic acid could also be detected. Ellagic acid has been described in unripe mango fruits for the first time by Saleh and El-Ansari (1974), whereas very recently Berardini et al. (2004) obtained no evidence for its presence even when using LC-MS for the detection of phenolic compounds. However,

the latter study mainly focussed on the characterization of gallotannins and benzophenone derivatives, which were present in abundance. It is therefore assumed that small amounts of ellagic acid could not be detected due to coelution with other polyphenolics. Another possible explanation for this finding is the release of ellagic acid from ellagitannins during pectin extraction in acidic milieu.

Based on a process recently developed for the recovery of pectin and phenolic compounds from apple pomace (Schieber, Hilt et al., 2003), the acidic peel extract was applied on a styrene–divinylbenzene copolymerisate resin (Process I), which has been approved for food use by the FDA. Preliminary laboratory-scale experiments had shown

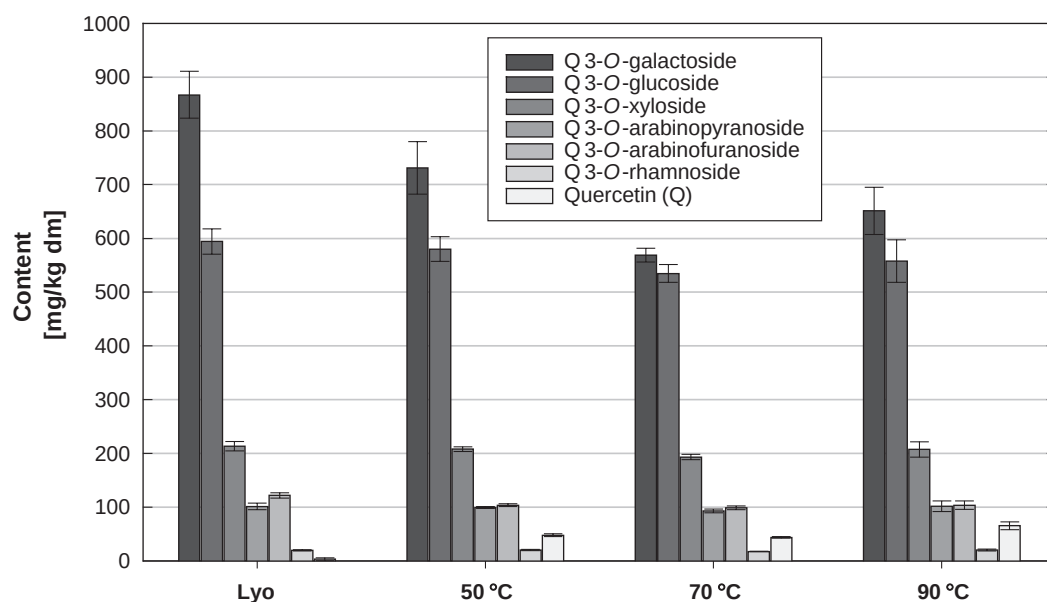


Fig. 4. Contents of quercetin and quercetin 3-*O*-glycosides in mango peels dried at 50, 70 and 90 °C, respectively. Lyo: lyophilized sample.

Table 2

L^* , a^* , b^* values, chroma (C^*) and hue angle (h°) of mango peel extract and mango peel pectin, respectively, before (–) and after (+) adsorptive removal of phenolic compounds ($n=2$)¹

Sample	Adsorption	L^*	a^*	b^*	C^*	h°
Mango peel extract	–	66.4±0.3	6.4±0.1	42.4±0.7	42.9±0.7	81.4±0.1
	+	80.3±0.1	1.1±0.1	24.3±0.1	24.3±0.0	87.4±0.0
Mango peel pectin	–	76.4±2.2	2.1±0.2	17.4±1.9	17.5±1.9	83.1±1.3
	+	78.1±0.6	2.3±0.2	15.4±0.2	15.6±0.1	81.5±1.0

¹Illuminant D65 and 10° observer angle.

that after application of 40 bed volumes of peel extract the concentration of phenolic compounds in the effluent increased. Therefore, the adsorption capacity of the resin for mango peel extract is comparable with the capacity for apple pomace extract because in that study similar amounts of extract were applied to the column (Schieber, Hilt et al., 2003). Color measurement data of the peel extract and the pectin obtained in Process I, respectively, before and after the adsorption of the phenolic compounds are shown in Table 2. As can be seen, L^* values and hue angle of the purified extract increased, whereas a^* , b^* and Chroma values decreased. These results indicate a decolorization of the peel extract during the adsorption process. In contrast, differences were not as pronounced for the pectins obtained from the extract before and after the adsorption of the phenolic compounds. Only a slight increase in the lightness and a slight decrease of the chroma value and hue angle were observed. Contrary to pectin obtained from crude apple pomace extracts, the mango peel pectin was not adversely affected by brown discoloration. Therefore, the effect of the adsorptive removal of phenolic compounds on the color of mango peel pectin was negligible. However, an improvement of the mango pectin color could nevertheless be important in industrial-scale mango pectin production, since even during short-term storage of the wet peels before drying oxidation of the phenolic compounds may occur, and refining of the pectin could become imperative.

Although no differences were observed with respect to the color of the pectins obtained before and after the adsorption process, their phenolic composition showed significant differences. As can be seen from Table 3, no flavonols and only small amounts of xanthones were detected in the pectin after the adsorption process, while a total amount of 270.9

mg/kg of these compounds was found in the pectin before the purification step. Pectin with a high content of phenolic compounds could be used as a functional ingredient, as previously suggested for mango peel fibres, which showed a high antioxidant activity (Larrauri, Gõni et al., 1996; Larrauri, Rupérez et al., 1996; Larrauri et al., 1997). On the other hand, browning of the pectin without removal of the phenolic compounds was observed during storage (over one month), while the pectin obtained after the adsorption procedure showed no discoloration, revealing that the undesired brown color is caused by oxidation and polymerization products of low molecular phenolics. The pectin yields were not significantly affected by the adsorption process. While the yield of the crude extract amounted to 17.6% based on peel dry matter, 16.4% were obtained from the purified extract.

As an alternative method for the combined recovery of pectin and phenolic compounds, a second process was developed, which is characterized by removal of alcohol from the precipitation bath, followed by application of the resulting aqueous solution on the adsorber resin (Process II). As described previously, color measurements of the pectin obtained in Process I had revealed that the adsorption of the phenolic compounds was not a prerequisite to produce a light-colored pectin. Therefore, the precipitation of the pectin was performed before the adsorptive removal of the polyphenols. As can be seen from Table 1, considerable amounts of the phenolic compounds remained in the aqueous phase of the precipitation bath. More than 70% of the flavonols and xanthones of the mango peel extract were recovered in the alcoholic solution after pectin precipitation. Preliminary experiments had shown that the recovery of a polyphenol-rich product could not be accomplished by evaporation and

Table 3

Contents (mg/kg dm) of flavonol and xanthone glycosides in mango peel pectin obtained before and after adsorptive removal of phenolic compounds ($n=2$)

	Mango peel pectin before polyphenol adsorption	Mango peel pectin after polyphenol adsorption
Mangiferin	216.0±12.3	36.9±4.5
Isomangiferin ¹	10.4±0.4	1.5±0.4
Mangiferin gallate ²	27.4±0.8	0.8±0.2
Isomangiferin gallate ²	8.0±0.4	0.2±0.1
Quercetin 3- <i>O</i> -glucoside	5.8±0.7	n.d.
Rhamnetin 3- <i>O</i> -galactoside/glucoside ³	0.6±0.0	n.d.
Quercetin	2.7±1.2	n.d.
Total	270.9±12.4	39.4±4.5

Quantified as ¹mangiferin ²mangiferin (including molecular weight correction factor) ³isorhamnetin 3-*O*-glucoside; n.d.: not detected.

lyophilization of the precipitation bath, because the high sugar contents impeded the drying of the residue and a sticky, viscous mass was obtained. Therefore, the solution was used for the adsorption process after evaporation of the organic solvent. Process II can easily be integrated in an existing industrial pectin production process, because the residue of the recycling process of the precipitation bath can be used for the adsorptive recovery of functional compounds.

The polyphenolic contents in the lyophilized eluates of both processes are shown in Table 4. While in total 129.4 mg/g polyphenols were detected in the lyophilizate obtained from Process I, only 71.0 mg/g dm could be found in the lyophilizate obtained from Process II. The profiles of the polyphenols were almost identical, revealing that during pectin precipitation there is no preferential adsorption of polyphenolic compounds to the pectin. Besides the predominant phenolic compound mangiferin, relatively large amounts of ellagic acid were detected. Compared with the amounts of ellagic acid in the peel extract and the precipitation bath (Table 1), this compound was accumulated during the adsorption processes. In previous investigations using apple pomace extracts, a lyophilizate with a total amount of phenolic compounds of 117.6 mg/g dm was obtained (Schieber, Hilt et al., 2003). The contents of phenolic compounds in extracts obtained from byproducts of industrial artichoke and cauliflower processing, using an Amberlite XAD 2 resin for purification, ranged from 98.9 to 242.8 mg/g dm and from 186.8 to 311.1 mg/g dm, respectively (Llorach, Espín, Tomás-Barberán, & Ferreres, 2002; Llorach, Espín, Tomás-Barberán, & Ferreres, 2003). Using the same procedure for byproducts of the industrial lettuce and chicory processing, the contents of phenolic compounds in the extracts ranged from 27.5 to 308.8 mg/g dm (Llorach, Tomás-Barberán, & Ferreres, 2004). Therefore, mango byproducts also represent a promising source for the recovery of polyphenolics.

The results of the rhodanine assay indicate that nearly all gallotannins were hydrolyzed during the extraction process. In the lyophilizate obtained from Process I 1.6% gallotannins (calculated as gallic acid) were determined, whereas only 0.8% gallotannins were found in the lyophilizate obtained

Table 5

Antioxidative capacity [$\mu\text{mol Trolox}/100\text{ g sample}$] of the lyophilizates and selected reference compounds ($n=2$)

	DPPH	TEAC	FRAP
Lyophilizate process I	487 $\pm$ 15	1139 $\pm$ 2	436 $\pm$ 8
Lyophilizate process II	443 $\pm$ 46	885 $\pm$ 6	361 $\pm$ 6
Gallic acid	1322 $\pm$ 144	3464 $\pm$ 168	1083 $\pm$ 59
Mangiferin	360 $\pm$ 71	1056 $\pm$ 49	352 $\pm$ 3
Quercetin 3- <i>O</i> -glucoside	370 $\pm$ 47	756 $\pm$ 37	319 $\pm$ 6

from Process II. Since the xanthone and benzophenone galates were also determined by the rhodanine assay, the real gallotannin content of the lyophilizates may be lower than indicated by the above mentioned values. Gallic acid, which was released during the hydrolysis of the gallotannins, was not retained by the resin and could therefore not be detected in the lyophilizate.

For the determination of the antioxidative capacity the DPPH, TEAC and FRAP assays were applied. The use of the TEAC assay modified according to Van den Berg et al. (1999, 2000) decreases the susceptibility to artifacts and prevents overestimation of antioxidative capacity. By generation of radicals before the addition of antioxidants, interference of compounds which affect radical formation is avoided (Sánchez-Moreno, 2002). The results showed that the polyphenol-rich lyophilizates displayed high antioxidative capacities as compared to mangiferin and quercetin 3-*O*-glucoside standard compounds, and may therefore be used as natural antioxidants or as functional food ingredients (Table 5). The lyophilizates obtained from Process I showed higher antioxidative capacities in all three assays. These findings indicate a correlation between the amount of phenolic compounds and the antioxidative capacity. The comparison with standard compounds gives evidence that the lyophilizates have a higher antioxidative capacity than mangiferin and quercetin 3-*O*-glucoside, indicating that the antioxidant capacity of the lyophilizates cannot only be explained by the presence of the xanthenes and flavonols. Consequently, the antioxidative capacity of ellagic acid and the benzophenone derivatives, which have not been included in this study, as well as of the galloyl moiety of the xanthenes and benzophenones also contribute to the total antioxidant activity of the samples. Furthermore, synergistic effects and other components such as Maillard products, for example, may also be considered.

4. Conclusions

Both processes presented in this paper are highly suitable for the combined recovery of pectin and phenolic compounds from industrial mango peel waste and may be integrated in the industrial pectin production. Because a high percentage of the phenolic compounds was degraded during the extraction of the peels, more investigations into the optimization of the extraction parameters are highly

Table 4

Composition [mg/g dm] of the phenolic fraction of the lyophilizates ($n=2$)

	Process I	Process II
Mangiferin	75.1 $\pm$ 1.3	43.6 $\pm$ 4.0
Isomangiferin ¹	3.0 $\pm$ 0.1	1.8 $\pm$ 0.1
Mangiferin gallate ²	7.6 $\pm$ 0.5	3.8 $\pm$ 0.2
Isomangiferin gallate ²	2.4 $\pm$ 0.3	1.3 $\pm$ 0.1
Quercetin 3- <i>O</i> -glucoside	1.8 $\pm$ 0.2	0.8 $\pm$ 0.0
Rhamnetin 3- <i>O</i> -galactoside/glucoside ³	0.2 $\pm$ 0.0	0.1 $\pm$ 0.0
Quercetin	2.7 $\pm$ 0.1	1.8 $\pm$ 0.1
Ellagic acid	36.6 $\pm$ 1.3	17.8 $\pm$ 1.5
Total	129.4 $\pm$ 2.0	71.0 $\pm$ 4.3

Quantified as ¹mangiferin ²mangiferin (including molecular weight correction factor) ³isorhamnetin 3-*O*-glucoside; n.d.: not detected.

desirable to improve the yield of polyphenols. Further investigations need to be directed at the application on the polyphenol rich lyophilizates as natural antioxidants and functional food or feed supplements.

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